

# Automatic Target-scoring Model based on Image Processing

Yuhua Li<sup>1</sup>, Chunyu Zhang<sup>1</sup>, Rui Cheng<sup>1</sup>, Yongan Xu<sup>1\*</sup>, Pu Li<sup>1</sup>, Huan Ma<sup>1</sup>

<sup>1</sup> Software Engineering College,  
Zhengzhou University of Light Industry,  
Zhengzhou 450001, China

\* Corresponding author: Yongan Xu, e-mail: 2018068@zzuli.edu.cn

**Abstract**—At present, in the field of military shooting, most of the target-scoring methods are artificial, acoustics-based, optics-based, and so on. These kinds of methods have their own disadvantages respectively such as high cost, time-consuming and low accuracy. To solve these problems, starting from the field of computer vision, combined with the image processing algorithms, in this paper, we propose an automatic target-scoring model. Firstly, image correction and subtraction operations are performed on the two images before and after shooting. Then, image processing operations such as binarization, opening and closing operations, and Canny edge detection are performed on the images that have eliminated most of the background interference, and a clear image of a single bullet hole is obtained. Finally, the Hough transform algorithm is used to identify the bullet holes, the Euclidean distance between the center of the bullet hole and the target surface is calculated, so the final target score is obtained. The testing images are all taken at a fixed angle of view 50 meters away from the targets. The experimental results show that the algorithm can accurately identify the location of bullet holes, with fast processing and more low cost than other algorithms.

**Keywords**- Automatic target-scoring; Image processing; Hough Transform

## I. INTRODUCTION

Target scoring is an important task for many applications includes national and international shooting competitions, military and police training academic, etc. [1]. Bullseye shooting is a shooting sports discipline where the shooter is standing in one place and shooting at a static paper target. The shooter's goal is to land as many hits in the center of the target as possible. [2] At present, military target training often uses chest ring targets as training samples for target shooting. Traditional target scoring methods often use artificial scoring, optoelectronic and laser systems, target-reading machines [3], acoustics-based scoring, fiber coding, two-layer electrode short-circuit sampling, etc. However, these target scoring methods have problems in terms of efficiency and cost. The artificial scoring method not only has a large workload, low efficiency, and potential safety hazards but also has large subjective factors, which are not conducive to fairness and justice. The Photoelectric positioning and the Fiber Coding have high precision, but they are easily interfered with by external light factors, and the later maintenance is difficult; The acoustics-based scoring has a long service life and is not disturbed by light compared with photoelectric positioning

scoring, but it is not suitable for small target surfaces and is easily disturbed by adjacent targets. The two-layer electrode short-circuit sampling has strong anti-interference and is not affected by external factors, but after each shooting, its metal layer is permanently damaged, and the cost is huge.

Recent years, with the development of computer vision, image processing based technologies have been widely used in the field of shooting training. Most image-based target scoring methods are composed of three parts: image preprocessing, target detection, and computational performance. The main technologies include the wavelet transform using fusion rules [4], the multi-threshold image segmentation [5] algorithms, template matching algorithms, etc. Among these methods, the wavelet transforms using fusion rules have a strong anti-interference ability, but its algorithm is complicated and time complexity is high, which cannot guarantee the timeliness of target scoring. The classical threshold image segmentation algorithm is simple and has high timeliness, but it has high requirements for light and is not suitable for environments with shadows and strong light. The template matching algorithm requires numerous bullet hole templates. However, in practice, the shape of the bullet holes is often irregular and the exact shape of the impact region, as detected by the computer software, is not always a perfect circle [6]. So, there are often missed detections during the identification process. At present, in the field of image processing, although the target scoring problem has been developed for many years, it is still in its infancy. The accuracy of target scoring is strongly dependent on external light factors, and its response time is also somewhat less than that of traditional methods.

Through research, this paper proposes an algorithm that combines image subtraction with generalized Hough transformation. Because the target of this experiment is not the entire target surface, it is the newly added bullet hole images in the target surface images after each shot. Therefore, to extract the only changing single bullet hole information on the target surface and reduce the noise interference of the entire target surface as much as possible, this paper uses the image subtraction method based on time series to process the two images before and after shooting. Due to external light interference and the oscillation of the bullet to the target surface during each target shooting, there is still a lot of noise in the image after image subtraction. In order to deal with these noises and identify bullet holes more accurately, this paper uses binarization, opening and closing operations,

median filtering, and Canny edge detection to sequentially process the images. After image preprocessing, a clear bullet hole can be obtained. Finally, the generalized Hough transformation is performed on the image to obtain the center coordinates of the bullet hole, the Euclidean distance from the center of the target surface is calculated, and the final target score is obtained. Compared with other algorithms based on image processing, the algorithm has low time complexity and can guarantee timeliness. Also, the algorithm is simple to implement and low in cost compared with the traditional target-scoring methods.

The second section of this paper introduces the existing target-scoring techniques. The third section explains the main algorithm and the principles used in image preprocessing and bullet hole recognition in detail. Section IV presents the experimental results and conducts experimental analysis. The last part is the conclusion of this paper.

## II. RELATED WORK

Since the invention of shooting weapons, the ability to report targets more conveniently and accurately has always been the focus of people's inventions and innovations. How can it be more convenient and more accurate? Since the 1980s, traditional electronics, acoustics, and optics have been gradually applied to the field of automatic target-scoring. The safety problems in artificial target-scoring have been solved successively, and the accuracy has also been improved to a certain extent, but there is a fatal problem of high cost. How can we guarantee accuracy at a lower cost? Since the 21st century, image processing technology has developed rapidly. The image captured by the camera only needs to be input into the system, the bullet hole information can be identified, and the target score can be calculated, thus solving the cost problem.

In 1989, C Sanctuary proposed an acoustics-based Scoring system for an air-to-ground strafing range is disclosed which

utilizes a pair of vector sensors for each target to pinpoint a target hit and/or miss by means of shell shock wave amplitude and time of arrival (TOA) differentials of the shock wave between the two sensors [7]. But, the acoustics-based scoring system is susceptible to shell shock waves from nearby targets. In 2005, KNESTEL Technologie & Elektronik GmbH from Germany designed shooting targets scanners called DISAG RM IV. This machine can scan the target surface fired by more than four different guns. However, this machine is very expensive.

With the rapid development of image processing and target detection, people began to use images to detect bullet holes. In 2012, Jacek Rudzinski and Marcin Luckner designed a Low-cost automatic scoring of shooting targets. This system requires no additional equipment and relies solely on existing straightforward image processing such as the Prewitt edge detection and the Hough transformation [8]. In 2016, Luo Jie proposed a bullet holes object detection based on wavelet transform using fusion rules [9]. By using wavelet transform technology, the method adopts different wavelet transform coefficients for images of different frequencies, thereby enhancing the identification of bullet holes in the image and reducing noise interference. With the gradual maturity of the convolutional neural network technology, in 2020, Zhu et al. published the article recognition of bullet holes based on video image analysis [10]. In this paper, the authors use the support vector machine algorithm to extract bullet hole information, pre-trains Faster R-CNN on the VOC2007 database, then use the bullet hole database established by the author for training, and finally use the trained network to detect the bullet hole location.

The above target-scoring algorithms based on image processing or deep learning have their own advantages in target scoring accuracy, implementation difficulty, and timeliness. Therefore, this paper takes the essence and gets rid of the dross, proposes an algorithm that ensures high timeliness and simplicity while ensuring accuracy.

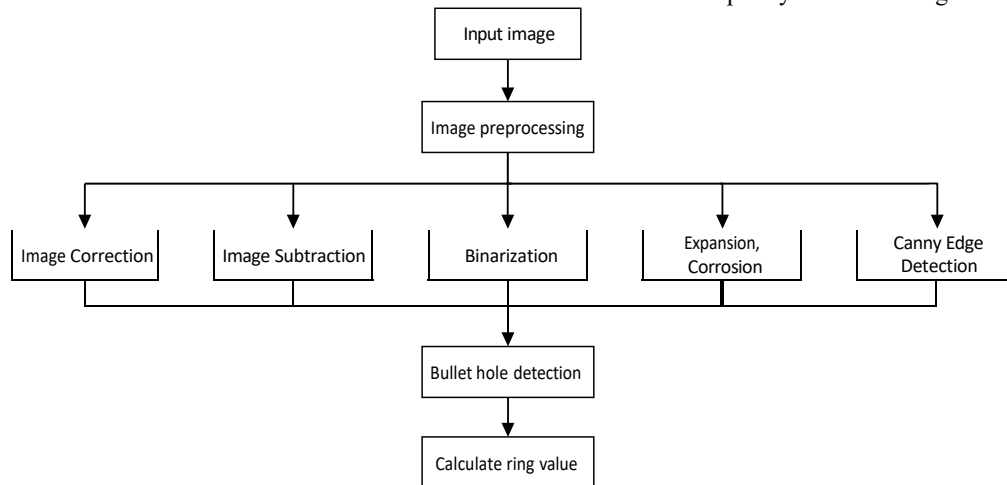


Figure 1. Flowchart of the proposed algorithm

## I. METHOD

The picture of the target surface captured by the camera generally contains a lot of interference information, such as the cluttered background and the interference of light and darkness. In order to eliminate these interferences, the algorithm firstly corrects the picture to eliminate the cluttered background interference and the target surface deformity caused by the shooting angle. Then performs image preprocessing on the target surface to eliminate the noise interference such as light, and highlight the bullet holes information. Finally, the Hough transformation is used to detect the bullet holes, and the results are calculated. The specific algorithm flow is shown in Figure 1.

### A. Image Preprocessing

#### 1) Image correction

Due to the posed problem of shooting, the input image often has geometric distortion. So, the image correction in the first step is very important. In this paper, the idea of perspective transformation is adopted, and four pixels are selected in the distorted image, which corresponds to the four pixels in the same position of the undistorted image. Then, the unknown coefficients in the mapping relationship are fitted by these four corresponding pixels, as the restoration basis for other pixels. The positions of the four pixels selected in this paper are shown in Figure 2.



Figure 2. The positions of the four pixels selected. In the figure, the four corners of the chest ring target surface are selected as the base coordinates of the perspective transformation.

The formula for perspective transformation is shown in Equation (1).

$$\begin{bmatrix} x^i & y^i & w^i \end{bmatrix} = \begin{bmatrix} u & v & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (1)$$

where  $u, v$  are the two-dimensional coordinates of the original image before perspective transformation,  $x^i, y^i$  and  $w^i$  are the three coordinates after perspective transformation. After perspective transformation, the pixel coordinates in the new image are shown in Equations (2) and (3) respectively.

$$x = \frac{x^i}{w^i} = \frac{a_{11} \times u + a_{12} \times v + a_{13} \times 1}{a_{31} \times u + a_{32} \times v + a_{33} \times 1} \quad (2)$$

$$y = \frac{y^i}{w^i} = \frac{a_{21} \times u + a_{22} \times v + a_{23} \times 1}{a_{31} \times u + a_{32} \times v + a_{33} \times 1} \quad (3)$$

Through the above formula, the corrected pixel point coordinates can be obtained. Finally, a square undistorted target surface image can be obtained.

Since difference between environmental factors and the nature of the input conversion device is taken into account, captured images by camera have color space distortion [11]. Therefore, before proceeding to the next step, the image needs to be uniformly converted into a grayscale image.

#### 2) Image subtraction

Image subtraction is the most important step of the algorithm in this paper. Image subtraction is to detect the moving objects from the difference between the existing frame and the reference frame [12]. After image correction, the image has been excluded from the interference of background and distortion. At this time, the newly added bullet hole is equivalent to a moving object. At present methods used in moving object detection are mainly the frame subtraction method, the background subtraction method and the optical flow method [13]. This paper uses frame subtraction to subtract the two pictures before and after the shooting, eliminates noise interference such as target ring lines and old bullet holes, and highlights the newly added bullet hole information.

The principle of image subtraction is very simple, that is, subtract the pixels in the same position of two images, and take the absolute value of the result to obtain the final image. The algorithm of image subtraction is as Formula (4), where  $I_1(i, j)$  is the image before shooting,  $I_2(i, j)$  is the image after shooting, and  $O(i, j)$  is the final image.

$$O(i, j) = \begin{cases} 0, & |I_1(i, j) - I_2(i, j)| < 0 \\ |I_1(i, j) - I_2(i, j)|, & |I_1(i, j) - I_2(i, j)| \geq 0 \end{cases} \quad (4)$$

#### 3) Binarization

The image aberration operation has filtered out most of the noise, and the grayscale histogram drawn at this time is shown in Figure 3. From this, we can see that after image subtraction, there are only two-pixel groups with grayscale distribution in the image, namely the background pixel group eliminated by image subtraction and the newly added bullet hole pixel group. Because the grayscale distribution of the image after the image subtraction operation is relatively regular, the bullet holes and the background each form a peak, so the threshold value can be taken at the valley. According to the actual experimental test, this algorithm selects 15 as the threshold for binarizing the image.

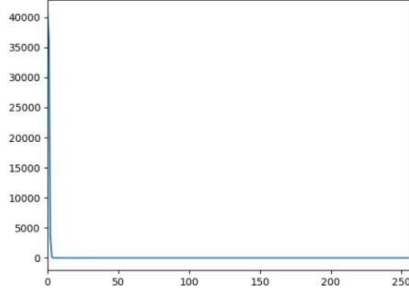


Figure 3. Grayscale histogram after image subtraction. The gray value in the grayscale image is divided into two parts: background and bullet holes.

#### 4) Opening and closing operations

Due to the impact force when the bullet penetrates the target surface, the bullet hole is often not a very regular circle, and there are generally many small cracks around the bullet hole, as shown in Figure 4. Since we finally use the Hough transformation to detect bullet holes, the rounder the bullet hole is, the more conducive it is to detect bullet holes.



Figure 4. Cracks around the bullet hole. Close-up shot of bullet holes on the target surface of the chest ring, the cracks on the target surface are clearly visible.

Before doing the opening and closing operations, we need to choose the right size kernel. The kernel size must be big enough to discriminate bullet hole from non-bullet hole, but not too big that it cannot detect novel bullet holes [14]. Therefore, in this paper, the  $5 \times 5$  kernel is used for the closing operation to fill in the gap of the outline, and then the  $3 \times 3$  kernel is used for the opening operation to remove the protruding burrs on the outline, followed that the small cracks bullet holes are eliminated as a whole. The algorithm for opening operation is as formula (5). The principle is to use the structural element B to corrode A first and then perform the expansion operation on the corrosion result with the same structural element.

$$A \circ B = (A \$ B) \oplus B \quad (5)$$

The algorithm of closing operation is as formula (6). The principle is to first use the structural element B to expand A, and then use the same structural element to perform the erosion operation on the expansion result. The process is just the opposite of the opening operation. After eliminating the interference, the surrounding of the bullet hole is very smooth, and the shape is more similar to a regular circle.

$$A \bullet B = (A \oplus B) \$ B \quad (6)$$

When people observe a picture, the visual focus is often the edge of the object in the picture, and the category and shape of the object are judged by the direction and composition of the edge. In a sense, the edge is the most obvious image feature. The purpose of edge detection is to find the set of pixels in the image whose brightness changes drastically. The so-called edge refers to the set of those pixels whose surrounding pixels have a step change in gray level. This paper uses the Canny edge detection. Canny's superior performance is due mainly to its provision of the ability to adjust the output quality by manipulating the edge detection parameters, Sigma and Threshold [15]. The main steps of Canny edge detection are as follows.

- a) Smooth the image with a Gaussian filter.
- b) Calculate gradient magnitude and direction using first-order partial derivative finite difference.
- c) Non-maximum suppression of gradient magnitudes.
- d) Detect and connect edges with a dual threshold algorithm.

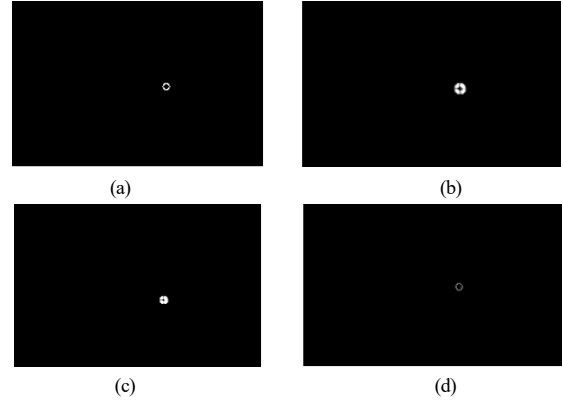


Figure 5. Various detection operator renderings. (a), (b), (c), (d) are the renderings of edge detection using Canny operator, Sobel operator, Schar operator, Laplacian operator respectively.

Canny edge detection is achieved by finding the local maximum value of the image gradient. This operator is a multi-stage optimization operator with filtering, enhancement and detection. Compared with the edge detection of other algorithms, as shown in Figure 5, Canny edge detection has higher accuracy, and the obtained boundary is single-pixel wide and the edge is clearer.

#### B. Bullet hole detection

In this paper, the generalized Hough transformation algorithm is used to detect bullet holes. The key idea of the Hough transform is to vote evidence from the image domain to the parametric domain, and then detect shapes in the parametric domain by identifying local-maximal responses. [16]. It achieves this by determining specific values of parameters which characterize these patterns. Spatially extended patterns are transformed so that they produce spatially compact features in a space of possible parameter values [17]. The generalized Hough transformation is more used to detect straight lines in practical applications, but it is also suitable for detecting any circle detection that can be

represented by  $(x, a) = 0$ . In an ideal situation, the center of the circle must lie on a line oriented normal to the edge direction. Therefore, we only have to move along the normal of every edge point to find the possible locations of centers. The distance between each edge point and the estimated center is a candidate for radius of the corresponding circle [18]. The standard Hough transformation makes use of this principle.

Since the parameter space of the traditional Hough transformation is three-dimensional, the time complexity is high, and the time and space required for detection are huge. Therefore, this algorithm adopts Hough transformation using image gradient information. This algorithm needs to differentiate the left and right sides of the standard equation of the circle with respect to  $x$ . At this time, the parameter radius  $r$  is eliminated, the parameter space becomes two-dimensional, and the calculation amount is greatly reduced. As shown in formula (7).

$$2(x - a) + 2(y - b) \frac{dy}{dx} = 0 \quad (7)$$

The Hough gradient algorithm is mainly divided into the following five steps:

a) First, the Sobel operator is used to calculate the original image in the  $x$  and  $y$  directions respectively, and the pixel gradient values in the  $x$  and  $y$  directions are obtained respectively. Then calculate the magnitude and direction of the gradient, and judge whether the image is circular according to the gradient direction. A circle if the gradient direction will point towards the center of the circle.

b) Set a two-dimensional counter  $N(i, j)$  at the center of the circle, initialized to all  $N(i, j) = 0$ .

c) Traverse all non-zero pixels in the Canny edge binary image and calculate its gradient direction. The gradient direction and the non-zero pixel point may correspond to a ray, the starting point is the non-zero pixel point, and the direction is the gradient direction. Increment the counter at the pixel where the ray passed.

d) According to the set threshold, when the counter is greater than the threshold, the point corresponding to the counter is considered to be the center of a circle. After the center of the circle is determined, the radius is gradually increased to find the circle.

After the above steps, the center coordinates of the bullet hole are obtained, and then the next score calculation can be performed.

### C. Grade calculation

By observing the military target surface, we found that although the target surface is human-shaped, it still follows the basic characteristics of a ground target, and the straight distance between each circle line is equal.

It can be seen from this that if the coordinates of the bullseye and the radius of the 10-ring are obtained, the ring value of the newly added single hole can be obtained by

calculation. The steps for calculating the grades in this paper are as follows.

a) Repeatedly use the opening and closing operations on an original target surface to eliminate other noise interference such as white ring lines and numbers, and obtain a binary image with only 10 ring areas left.

b) Perform Hough transformation on the binary image, detect a circular area with 10 rings, obtain the coordinates of the center of the circle  $(x, y)$ , and the radius  $r$ .

c) Calculate the Euclidean distance  $l$  between the coordinates of the center of the circle  $(x, y)$  and the coordinates of the bullet hole.

d) According to the radius, judge the ring value and output the result.

## II. EXPERIMENT

The target surface used in the experiment in this paper is a new type of target surface. Compared with the ordinary paper target surface, the target surface does not have the problem of incompatibility between the target paper and the shooting board, which improves the detection accuracy. The image data used in this experiment are all field shots from the army shooting site to ensure the authenticity and reliability of the experiment.

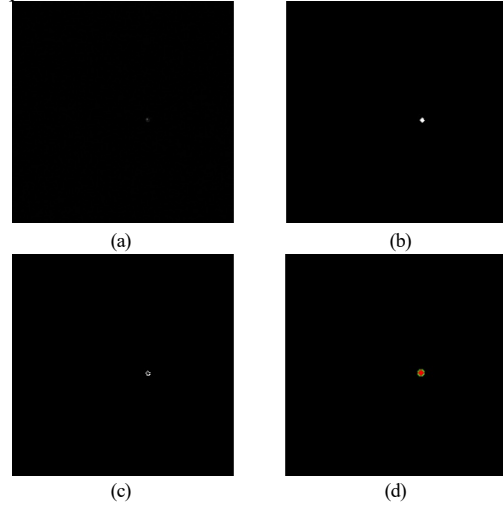


Figure 6. After correcting the image acquired by the camera and image subtraction operation, (a) is obtained Experimental renderings. After performing various image processing operations on (a), we obtained (b), in which we can see clear round bullet holes. Use Canny edge detection on image (b) to get image (c). Finally, the Hough transformation is used to identify bullet holes to obtain the image (d).

Figure 6 is the effect diagram of the overall experimental process of this experiment. As shown in Figure 6(a), image subtraction, as the central idea of this algorithm, can indeed eliminate huge noise interference and focus image processing on bullet holes. As can be seen from Figure 6(d), this algorithm can accurately identify the location of bullet holes by using the Hough transformation.

For obvious comparison, the images with an interval of 4 shots are selected as the target images of the template



matching method and this algorithm. For this image, the template detection algorithm and the algorithm in this paper are used to detect bullet holes. Figure 7(a) is the bullet hole detection diagram based on the template matching algorithm, and Figure 7(b) is the bullet hole detection diagram using the algorithm in this paper. The disadvantage of the template matching method is that enough bullet hole templates must be collected for detection, so as to ensure the accuracy of detection. However, the impact strength of the bullet and the target surface is different, and the material of the target surface paper is different, and the bullet holes are often strange. It is difficult to guarantee that the collected bullet hole templates will contain all types of bullet holes. It can be seen that the algorithm in this paper is better than the template matching algorithm in terms of bullet hole detection accuracy.

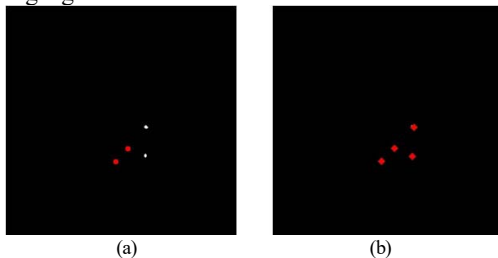


Figure 7. Bullet hole detection comparison. (a) is the bullet hole identification diagram of the template matching algorithm. (b) is the bullet hole detection map of the algorithm in this paper

After a lot of experiments, the results show that the algorithm can accurately identify bullet holes and report the target correctly. In a period of 100 rounds of continuous shooting, the accuracy of the algorithm in identifying bullet holes can reach 92.00%, and each identification takes an average of 867ms. This shows that the algorithm has high accuracy and real-time performance.

### III. CONCLUSION

In this paper, an automatic target-scoring model based on image processing is proposed. Image preprocessing techniques such as image correction, opening and closing operations, and image subtraction are used to eliminate target surface noise. Then, traditional image detection technology such as Hough transformation is used to detect the center coordinates of the bullet hole and the center coordinates of the target surface. Finally, the Euclidean distance and the characteristics of the circular target are used to calculate the final ring value. The proposed algorithm is simple and compared with traditional automatic target-scoring and other image-based target-scoring technologies, the timeliness and accuracy are higher. However, this algorithm still has a lot of room for improvement in the processing of anti-interference ability.

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